

Origin of Species that the facts he presented were not necessarily original; it was the consequences that he thought were “interesting” (as he put it with characteristic Victorian modesty). In the end it is those who derive consequences and seize the importance of the ideas, seeing their real value, who win the day. They are the ones who can talk about the subject.

So let me describe Mandelbrotian geometry.

The Geometry of Nature

Triangles, squares, circles, and the other geometric concepts that made many of us yawn in the classroom may be beautiful and pure notions, but they seem more present in the minds of architects, design artists, modern art buildings, and schoolteachers than in nature itself. That’s fine, except that most of us aren’t aware of this. Mountains are not triangles or pyramids; trees are not circles; straight lines are almost never seen anywhere. Mother Nature did not attend high school geometry courses or read the books of Euclid of Alexandria. Her geometry is jagged, but with a logic of its own and one that is easy to understand.

I have said that we seem naturally inclined to Platonify, and to think exclusively in terms of studied material: nobody, whether a bricklayer or a natural philosopher, can easily escape the enslavement of such conditioning. Consider that the great Galileo, otherwise a debunker of falsehoods, wrote the following:

The great book of Nature lies ever open before our eyes and the true philosophy is written in it. ... But we cannot read it unless we have first learned the language and the characters in which it is written. ... It is written in mathematical language and the characters are triangles, circles and other geometric figures.

Was Galileo legally blind? Even the great Galileo, with all his alleged independence of mind, was not capable of taking a clean look at Mother Nature. I am confident that he had windows in his house and that he ventured outside from time to time: he should have known that triangles are not easily found in nature. We are so easily brainwashed.

We are either blind, or illiterate, or both. That nature’s geometry is not Euclid’s was so obvious, and nobody, almost nobody, saw it.

This (physical) blindness is identical to the ludic fallacy that makes us think casinos represent randomness.

Fractality

But first, a description of fractals. Then we will show how they link to what we call power laws, or scalable laws.